

UNIT 4

Special Continuous Distributions

[Normal, Gamma and Exponential]

Normal Distribution

[Also known as the **Gaussian distribution** , in honor of Karl Friedrich Gauss (1777–1855)]

A continuous random variable X having the bell-shaped distribution of Figure is called a **normal random variable**. The mathematical equation for the probability distribution of the normal variable depends on the two parameters μ and σ , its **mean** and **standard deviation**, respectively. Hence, we denote the values of the density of X by $n(x; \mu, \sigma)$.

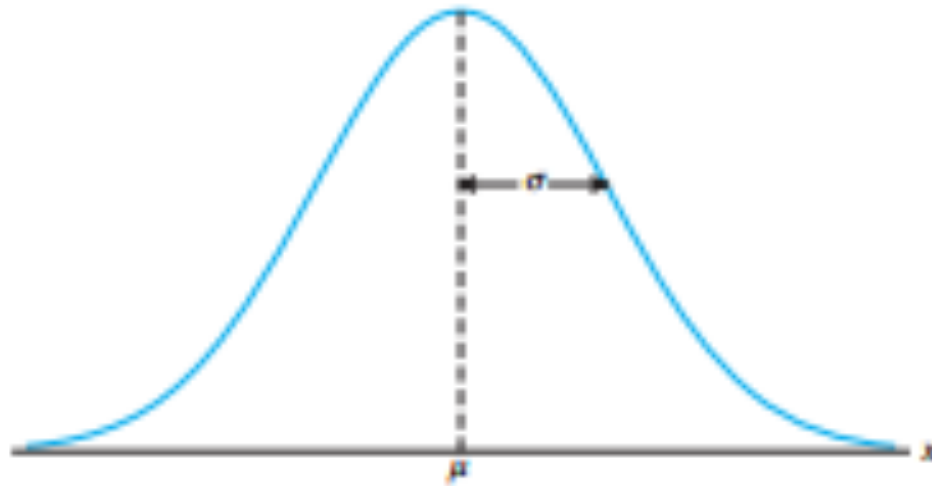


Figure 1.1 : The normal curve.

Probability density function of normal distribution

The density of the normal random variable X , with mean μ and variance σ^2 , is

$$n(x; \mu, \sigma) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{1}{2\sigma^2}(x-\mu)^2}, \quad -\infty < x < \infty,$$

where $\pi = 3.14159\dots$ and $e = 2.71828\dots$

Properties of Normal distribution

1. The mode, which is the point on the horizontal axis where the curve is a maximum, occurs at $x = \mu$.
2. The curve is symmetric about a vertical axis through the mean μ .
3. The curve has its points of inflection at $x = \mu \pm \sigma$; it is concave downward if $\mu - \sigma < X < \mu + \sigma$ and is concave upward otherwise.
4. The normal curve approaches the horizontal axis asymptotically as we proceed in either direction away from the mean.
5. The total area under the curve and above the horizontal axis is equal to 1.
6. The mean and variance of $n(x; \mu, \sigma)$ are μ and σ^2 , respectively. Hence, the standard deviation is σ .

Standard Normal variate

The distribution of a normal random variable with mean 0 and variance 1 is called a ***standard normal distribution***.

This can be done by means of the transformation

$$Z = \frac{X - \mu}{\sigma}.$$

The probability density function of Z is

$$f(z) = \frac{1}{\sqrt{2\pi}} e^{-\frac{z^2}{2}}; -\infty < Z < \infty$$

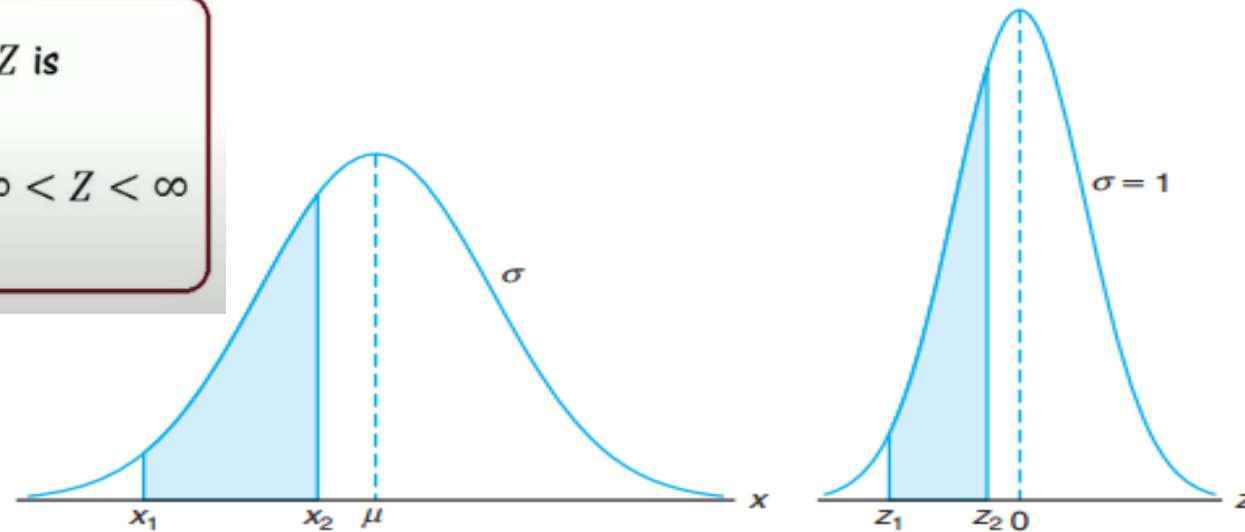


Figure 1.1: The original and transformed normal distributions.

Moment generating function (m.g.f) of Normal Distribution

- The moment generating function of normal distribution ;

$$M_X(t) = e^{\mu t + \frac{1}{2}\sigma^2 t^2}$$

MGF for standard random variable $Z \sim N(0,1)$

$$M_Z(t) = e^{\frac{t^2}{2}}$$

Find mean and Variance using MGF

Example 1

6.5 Given a standard normal distribution, find the area under the curve that lies

- (a) to the left of $z = -1.39$;
- (b) to the right of $z = 1.96$;
- (c) between $z = -2.16$ and $z = -0.65$;
- (d) to the left of $z = 1.43$;
- (e) to the right of $z = -0.89$;
- (f) between $z = -0.48$ and $z = 1.74$.

Example 2

6.8 Given a normal distribution with $\mu = 30$ and $\sigma = 6$, find

- (a) the normal curve area to the right of $x = 17$;
- (b) the normal curve area to the left of $x = 22$;
- (c) the normal curve area between $x = 32$ and $x = 41$;
- (d) the value of x that has 80% of the normal curve area to the left;
- (e) the two values of x that contain the middle 75% of the normal curve area

Example 3

Let X is normally distributed and mean of X is 12 and standard deviation is 4. Find out the probability of the following

a) $X \geq 20$, b) $X \leq 20$ and c) $0 \leq X \leq 12$.

Sol:

Example 4

6.11 A soft-drink machine is regulated so that it discharges an average of 200 milliliters per cup. If the amount of drink is normally distributed with a standard deviation equal to 15 milliliters,

- (a) what fraction of the cups will contain more than 224 milliliters?
- (b) what is the probability that a cup contains between 191 and 209 milliliters?
- (c) how many cups will probably overflow if 230-milliliter cups are used for the next 1000 drinks?
- (d) below what value do we get the smallest 25% of the drinks?

Example 5

6.6 Find the value of z if the area under a standard normal curve

- (a) to the right of z is 0.3622;
- (b) to the left of z is 0.1131;
- (c) between 0 and z , with $z > 0$, is 0.4838;
- (d) between $-z$ and z , with $z > 0$, is 0.9500

PRACTICE QUESTIONS

6.7 Given a standard normal distribution, find the value of k such that

(a) $P(Z > k) = 0.2946$;

(b) $P(Z < k) = 0.0427$;

(c) $P(-0.93 < Z < k) = 0.7235$.

Given that X has a normal distribution with $\mu = 300$ and $\sigma = 50$, find the probability that X assumes a value greater than 362.

PRACTICE QUESTIONS

6.9 Given the normally distributed variable X with mean 18 and standard deviation 2.5, find

- (a) $P(X < 15)$;
- (b) the value of k such that $P(X < k) = 0.2236$;
- (c) the value of k such that $P(X > k) = 0.1814$;
- (d) $P(17 < X < 21)$

Let X is normally distributed and mean of X is 30 and standard deviation is 5. Find out the probability of the following

- a) $26 \leq X \leq 40$ b) $X \geq 45$ and c) $|X - 30| > 5$.

PRACTICE QUESTIONS

6.13 A research scientist reports that mice will live an average of 40 months when their diets are sharply restricted and then enriched with vitamins and proteins.

Assuming that the lifetimes of such mice are normally distributed with a standard deviation of 6.3 months, find the probability that a given mouse will live

- (a) more than 32 months;
- (b) less than 28 months;
- (c) between 37 and 49 months

PRACTICE QUESTIONS

6.18: The heights of 1000 students are normally distributed with a mean of 174.5 centimeters and a standard deviation of 6.9 centimeters. Assuming that the heights are recorded to the nearest half-centimeter, how many of these students would you expect to have heights

- (a) less than 160.0 centimeters?
- (b) between 171.5 and 182.0 centimeters inclusive?
- (c) equal to 175.0 centimeters?
- (d) greater than or equal to 188.0 centimeters?

PRACTICE QUESTIONS

6.21 The tensile strength of a certain metal component is normally distributed with a mean of 10,000 kilograms per square centimeter and a standard deviation of 100 kilograms per square centimeter. Measurements are recorded to the nearest 50 kilograms per square centimeter.

(a) What proportion of these components exceed 10,150 kilograms per square centimeter in tensile strength?

(b) If specifications require that all components have tensile strength between 9800 and 10,200 kilograms per square centimeter inclusive, what proportion of pieces would we expect to scrap?

PRACTICE QUESTIONS

- 6.22** If a set of observations is normally distributed, what percent of these differ from the mean by
- (a) more than 1.3σ ?
 - (b) less than 0.52σ ?

An electrical firm manufactures light bulbs that have a life, before burn-out, that is normally distributed with mean equal to 800 hours and a standard deviation of 40 hours. Find the probability that a bulb burns between 778 and 834 hours.

Sol:

Normal Approximation to the Binomial

If X is a binomial random variable with mean $\mu = np$ and variance $\sigma^2 = npq$, then the limiting form of the distribution of

$$Z = \frac{X - np}{\sqrt{npq}},$$

as $n \rightarrow \infty$, is the standard normal distribution $n(z; 0, 1)$.

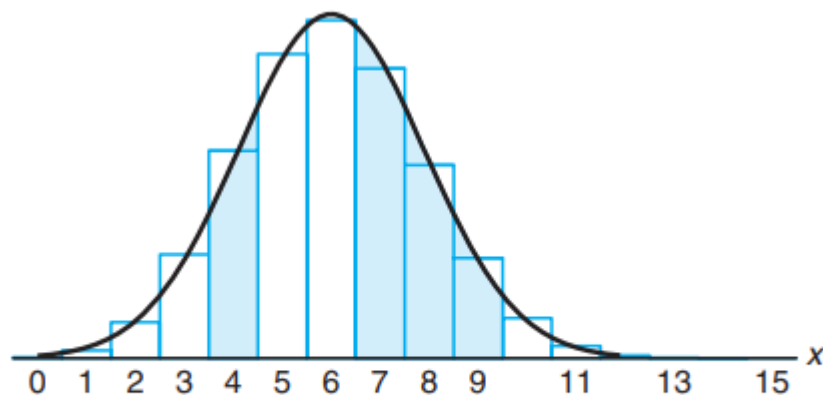


Figure 6.23: Normal approximation of $b(x; 15, 0.4)$ and $\sum_{x=7}^9 b(x; 15, 0.4)$.

if we seek the area under the normal curve to the left of, say, x , it is more accurate to use $x + 0.5$. This is a correction to accommodate the fact that a discrete distribution is being approximated by a continuous distribution. The correction **+0.5** is called a **continuity correction**.

Example 1

6.27 The probability that a patient recovers from a delicate heart operation is 0.9. of the next 100 patients having this operation, what is the probability that

(a) between 84 and 95 inclusive survive?

(b) fewer than 86 survive?

Example 2

A multiple-choice quiz has 200 questions, each with 4 possible answers of which only 1 is correct. What is the probability that sheer guesswork yields from 25 to 30 correct answers for the 80 of the 200 problems about which the student has no knowledge?

PRACTICE QUESTIONS

6.29 If 20% of the residents in a U.S. city prefer a white telephone over any other color available, what is the probability that among the next 1000 telephones installed in that city

(a) between 170 and 185 inclusive will be white?

(b) at least 210 but not more than 225 will be white?

The probability that a patient recovers from a rare blood disease is 0.4. If 100 people are known to have contracted this disease, what is the probability that fewer than 30 survive?

If 20% of the residents in an Indian city prefer a white telephone over any other color available, what is the probability that among the next 1000 telephones installed in that city (a) between 170 and 185 inclusive will be white? (b) at least 210 but not more than 225 will be white?

MCQS on Normal Distributions

1. **Standard normal distribution has which of the following properties**

- a) Mean=Variance=1
- b) Standard deviation=Variance=0
- c) Mean=0, Variance=1
- d) Mean=standard deviation

2. **If $X \sim N(5, 1)$ the probability density function for the normal variate X is**

- a) $\frac{1}{5\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-1}{5}\right)^2}$ b) $\frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-1}{5}\right)^2}$ c) $\frac{1}{5\sqrt{2\pi}} e^{-\frac{1}{2}x^2}$ d) $\frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}(x-1)^2}$

3. **If m.g.f of normal variate is $M_X(t) = e^{\left(\frac{24t+16t^2}{2}\right)}$ then *mean(μ) and variance(σ^2) of X are***

- a) $\mu = 24$ $\sigma^2 = 4$ b) $\mu = 12$ $\sigma^2 = 4$
c) $\mu = 12$ $\sigma^2 = 16$ d) $\mu = 24$ $\sigma^2 = 16$

4. **If $X \sim N(65, 25^2)$ with mean and variance as parameters, then value of standardized normal variable for $X = 35$ is**

- (A) -1.8 (B) -1.6 (C) -1.4 (D) -1.2

MCQS on Normal Distributions (Gate exam)

1. A normal random variable X has the following probability density function

$$f(x) = \frac{1}{\sqrt{8\pi}} e^{-\left[\frac{(x-1)^2}{8}\right]}, \quad -\infty < x < \infty. \text{ Then } \int_1^{\infty} f(x) dx =$$

- a) 0 b) $\frac{1}{2}$ c) $1 - \frac{1}{e}$ d) 1

2. Let X be a normal random variable with mean zero and variance 9. If $a = P(X \geq 3)$, then $P(|X| \leq 3)$ equals to

- a) a b) $2a$ c) $1-2a$ d) $1-a$

3. If X is a normal variate with mean 30 and standard deviation 5, what is the probability $P(26 \leq X \leq 34)$?

Given $A(Z > 0.8) = 0.2118$

- a) 0.2118 b) 0.4236 c) 0.7882 d) 0.5764

4. Let X be a random variable following normal distribution with mean +1 and variance 4. Let Y be another normal variable with mean -1 and variance unknown.

If $P(X \leq -1) = P(Y \geq 2)$, the standard deviation of Y is

- a) 3 b) 2 c) $\sqrt{2}$ d) 1

Gamma Distribution

- **Continuous** distribution
- R.v is always positive i.e defined for X in $(0, \infty)$
- Defined as **Skewed distribution**
- Occurs as **waiting time** between **events**
- Also **predict** the wait time until future event occur
- Predicts the **wait time** until the k^{th} **event** occurs

Gamma Distribution

Gamma Distribution The continuous random variable X has a **gamma distribution**, with parameters α and β , if its density function is given by

$$f(x; \alpha, \beta) = \begin{cases} \frac{1}{\beta^\alpha \Gamma(\alpha)} x^{\alpha-1} e^{-x/\beta}, & x > 0, \\ 0, & \text{elsewhere,} \end{cases}$$

where $\alpha > 0$ and $\beta > 0$.

The gamma function is defined by

$$\Gamma(\alpha) = \int_0^\infty x^{\alpha-1} e^{-x} dx, \quad \text{for } \alpha > 0.$$

Some properties of gamma function

$$\Gamma(\alpha) = (\alpha - 1)\Gamma(\alpha - 1).$$

$$\Gamma(n) = (n - 1)! \text{ for a positive integer } n.$$

$$\Gamma(1) = 1. \quad \Gamma(1/2) = \sqrt{\pi}.$$

Mean, Variance and m.g.f of Gamma Distribution

The mean and variance of the gamma distribution are

$$\mu = \alpha\beta \text{ and } \sigma^2 = \alpha\beta^2.$$

The moment generating function of gamma distribution is

$$M_X(t) = (1 - t\beta)^{-\alpha}$$

Find mean and Variance using MGF

Example 1

In a certain city, the daily consumption of electric power, in millions of kilowatt-hours, is a random variable X having a gamma distribution with mean $\mu = 6$ and variance $\sigma^2 = 12$.

(a) Find the values of α and β . (b) Find the probability that on any given day the daily power consumption will exceed 12 million kilowatt hours.

Example 2

6.41 If a random variable X has the gamma distribution with $\alpha = 2$ and $\beta = 1$, find $P(1.8 < X < 2.4)$.

Example 3

Suppose that telephone calls arriving at a particular switchboard follow a Poisson process with an average of 5 calls coming per minute. What is the probability that up to a minute will elapse by the time 2 calls have come in to the switchboard?

Example 4

6.42 Suppose that the time, in hours, required to repair a heat pump is a random variable X having a gamma distribution with parameters $\alpha = 2$ and $\beta = 1/2$. What is the probability that on the next service call

- (a) at most 1 hour will be required to repair the heat pump?
- (b) at least 2 hours will be required to repair the heat pump?

MCQs on Gamma Distributions

1. The value of $\Gamma(\frac{3}{2})=$

- a) $\frac{\sqrt{\pi}}{2}$ b) $\frac{\sqrt{\pi}}{3}$ c) $\frac{3\sqrt{\pi}}{2}$ d) none

2. If mean and variance of the gamma distribution is given as 24 and 192 respectively. Then the values of parameters α (shape) and β (scale) are

- a) $\alpha = 8$, $\beta = 3$
b) $\alpha = 3$, $\beta = 8$
c) $\alpha = 9$, $\beta = 8$
d) none

3. If random variable X has a Gamma distribution with $\alpha = 2$ and $\beta = 3$. Then the mean and standard deviation (S.D) of X are

- a) mean= 6 , S.D.= 18
b) mean= 18 , S.D.= 6
c) mean= 6 , S.D.= 4.2426
d) mean=12, S.D.=4.2426

4. On Sunday morning, customers arrive at cafeteria follows a Poisson process with average rate 15 customers per hour. What is the average amount of time elapse before 3 customers arrive in cafeteria.

- a) 4 mins b) 9 mins c) 12 mins d) 16mis

PRACTICE QUESTIONS

6.44 In a certain city, the daily consumption of electric power, in millions of kilowatt hours, is a random variable X having a gamma distribution with mean $\mu = 6$ and variance $\sigma^2 = 12$.

(a) Find the values of α and β .

(b) Find the probability that on any given day the daily power consumption will exceed 12 million kilowatt hours.

In a biomedical study with rats, a dose-response investigation is used to determine the effect of the dose of a toxicant on their survival time. The toxicant is one that is frequently discharged into the atmosphere from jet fuel. For a certain dose of the toxicant, the study determines that the survival time, in weeks, has a gamma distribution with $\alpha = 5$ and $\beta = 10$. What is the probability that a rat survives no longer than 60 weeks?

Following figure showing graphs of several gamma distributions for certain specified values of the parameters α and β .

The special gamma distribution for which $\alpha = 1$ is called the exponential distribution.

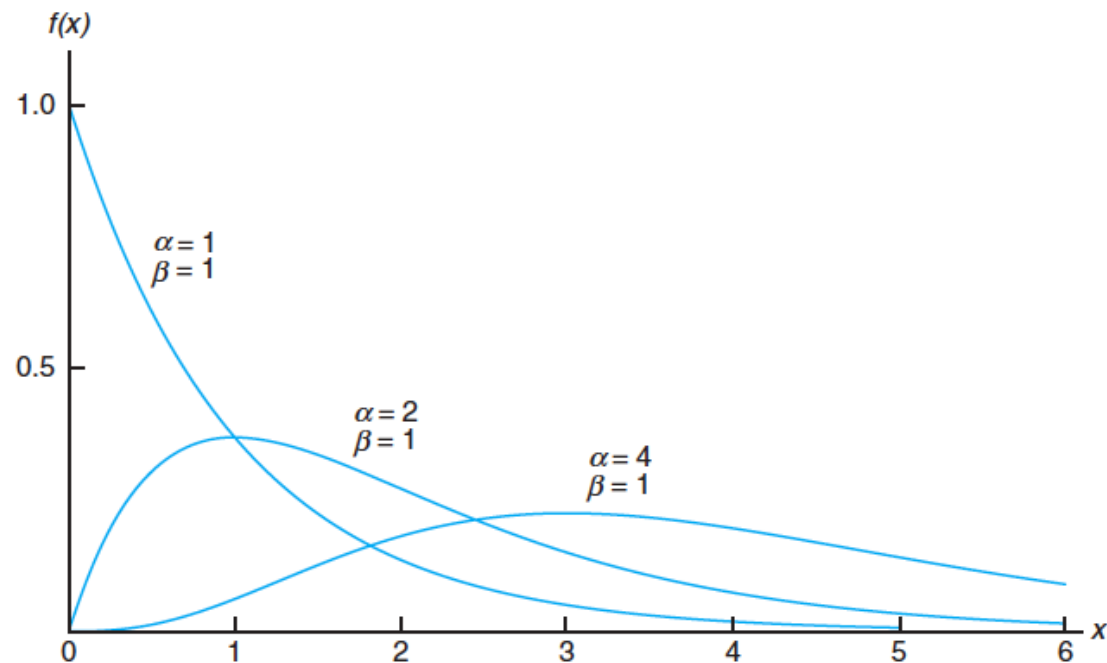


Figure 6.28: Gamma distributions.

Exponential distribution

Exponential Distribution

The continuous random variable X has an **exponential distribution**, with parameter β , if its density function is given by

$$f(x; \beta) = \begin{cases} \frac{1}{\beta} e^{-x/\beta}, & x > 0, \\ 0, & \text{elsewhere,} \end{cases}$$

where $\beta > 0$.

Mean, variance and m.g.f. of exponential distribution

The mean and variance of the exponential distribution are

$$\mu = \beta \text{ and } \sigma^2 = \beta^2.$$

The moment generating function of exponential distribution is

$$M_X(t) = (1 - t\beta)^{-1}$$

Find mean and Variance using MGF

Remarks-

- The **time** between **events** in a **Poisson process** is an **exponential** distribution
- Poisson- Event per single unit of time
- **Exponential – Time per single event**
- **The time required for first event to occur is exponential distribution**
- The **mean** of **exponential distribution** is a parameter **beta**, the **reciprocal** of the **parameter** in the **Poisson** distribution
- **Beta** is called mean time between events
- The **time** until the **occurrence** of **Poisson event** is random variable described by **exponential** distribution whereas the **time** until a **specified number** of **Poisson events** occur is a random variable described by **gamma** distribution
- The **specific number** of **events** is the parameter **alpha** in the gamma distribution

Example 1

6.81 The length of time between breakdowns of an essential piece of equipment is important in the decision of the use of auxiliary equipment. An engineer thinks that the best model for time between breakdowns of a generator is the exponential distribution with a mean of 15 days.

- (a) If the generator has just broken down, what is the probability that it will break down in the next 21 days?
- (b) What is the probability that the generator will operate for 30 days without a breakdown?

Example 2

A light bulb manufacturer claims that their bulbs have an average lifetime of 800 hours. Calculate the probability that a light bulb lasts longer than 1000 hours.

PRACTICE QUESTIONS

6.45 The length of time for one individual to be served at a cafeteria is a random variable having an exponential distribution with a mean of 4 minutes. What is the probability that a person is served in less than 3 minutes on at least 4 of the next 6 days?

Suppose that a study of a certain computer system reveals that the response time, in seconds, has an exponential distribution with a mean of 3 seconds.

- (a) What is the probability that response time exceeds 5 seconds?
- (b) What is the probability that response time exceeds 10 seconds?

PRACTICE QUESTIONS

The lifetime of a certain type of battery follows an Exponential distribution with a mean of 500 hours. What is the probability that a battery lasts less than 400 hours?

A machine has a failure rate of 0.2 failures per hour. What is the probability that the machine will operate without a failure for at least 5 hours?

Suppose that a study of a certain computer system reveals that the response time, in seconds, has an exponential distribution with a mean of 3 seconds.

MCQs on Exponential distributions

1. **If a continuous random variable X has an exponential distribution with mean 0.5. Then $P(X > 1/2) =$**
a) e^{-1} b) $e^{-1/2}$ c) $(1/2) e^{-1}$ d) $[1/2] e^{-1/2}$
2. **Suppose that a system contains a certain type of component whose lifetime is T years. The random variable T is modelled nicely by the exponential distribution with mean time to failure is $\beta = 5$.
The probability that a given component installed on a system is still working after 8 years is**
a) 0.16 b) 0.17 c) 0.18 d) 0.2

3. **A continuous random variable X has probability density function given by**

$$f(x) = \begin{cases} 2e^{-2x}, & x \geq 0 \\ 0, & \text{otherwise} \end{cases}$$

The mean and variance of X are

- a) $\frac{1}{2}, 1/8$ b) $\frac{1}{2}, \frac{1}{4}$ c) $1, \frac{1}{2}$ d) $2, \frac{1}{2}$

3. **The lifetime T of an alkaline battery is exponentially distributed with $\theta = 0.05$ per hour.
What are mean and standard deviation of batteries lifetime?**

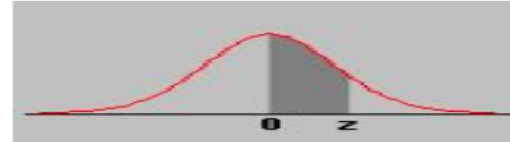
- a) mean=10, S.D.=20 c) mean=20, S.D.=10
b) mean=20, S.D.=20 d) mean=10, S.D.=10



Table A.3 Areas under the Normal Curve

<i>z</i>	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
−3.4	0.0003	0.0003	0.0003	0.0003	0.0003	0.0003	0.0003	0.0003	0.0003	0.0002
−3.3	0.0005	0.0005	0.0005	0.0004	0.0004	0.0004	0.0004	0.0004	0.0004	0.0003
−3.2	0.0007	0.0007	0.0006	0.0006	0.0006	0.0006	0.0006	0.0005	0.0005	0.0005
−3.1	0.0010	0.0009	0.0009	0.0009	0.0008	0.0008	0.0008	0.0008	0.0007	0.0007
−3.0	0.0013	0.0013	0.0013	0.0012	0.0012	0.0011	0.0011	0.0011	0.0010	0.0010
−2.9	0.0019	0.0018	0.0018	0.0017	0.0016	0.0016	0.0015	0.0015	0.0014	0.0014
−2.8	0.0026	0.0025	0.0024	0.0023	0.0023	0.0022	0.0021	0.0021	0.0020	0.0019
−2.7	0.0035	0.0034	0.0033	0.0032	0.0031	0.0030	0.0029	0.0028	0.0027	0.0026
−2.6	0.0047	0.0045	0.0044	0.0043	0.0041	0.0040	0.0039	0.0038	0.0037	0.0036
−2.5	0.0062	0.0060	0.0059	0.0057	0.0055	0.0054	0.0052	0.0051	0.0049	0.0048
−2.4	0.0082	0.0080	0.0078	0.0075	0.0073	0.0071	0.0069	0.0068	0.0066	0.0064
−2.3	0.0107	0.0104	0.0102	0.0099	0.0096	0.0094	0.0091	0.0089	0.0087	0.0084
−2.2	0.0139	0.0136	0.0132	0.0129	0.0125	0.0122	0.0119	0.0116	0.0113	0.0110
−2.1	0.0179	0.0174	0.0170	0.0166	0.0162	0.0158	0.0154	0.0150	0.0146	0.0143
−2.0	0.0228	0.0222	0.0217	0.0212	0.0207	0.0202	0.0197	0.0192	0.0188	0.0183
−1.9	0.0287	0.0281	0.0274	0.0268	0.0262	0.0256	0.0250	0.0244	0.0239	0.0233
−1.8	0.0359	0.0351	0.0344	0.0336	0.0329	0.0322	0.0314	0.0307	0.0301	0.0294
−1.7	0.0446	0.0436	0.0427	0.0418	0.0409	0.0401	0.0392	0.0384	0.0375	0.0367
−1.6	0.0548	0.0537	0.0526	0.0516	0.0505	0.0495	0.0485	0.0475	0.0465	0.0455
−1.5	0.0668	0.0655	0.0643	0.0630	0.0618	0.0606	0.0594	0.0582	0.0571	0.0559
−1.4	0.0808	0.0793	0.0778	0.0764	0.0749	0.0735	0.0721	0.0708	0.0694	0.0681
−1.3	0.0968	0.0951	0.0934	0.0918	0.0901	0.0885	0.0869	0.0853	0.0838	0.0823
−1.2	0.1151	0.1131	0.1112	0.1093	0.1075	0.1056	0.1038	0.1020	0.1003	0.0985
−1.1	0.1357	0.1335	0.1314	0.1292	0.1271	0.1251	0.1230	0.1210	0.1190	0.1170
−1.0	0.1587	0.1562	0.1539	0.1515	0.1492	0.1469	0.1446	0.1423	0.1401	0.1379
−0.9	0.1841	0.1814	0.1788	0.1762	0.1736	0.1711	0.1685	0.1660	0.1635	0.1611
−0.8	0.2119	0.2090	0.2061	0.2033	0.2005	0.1977	0.1949	0.1922	0.1894	0.1867
−0.7	0.2420	0.2389	0.2358	0.2327	0.2296	0.2266	0.2236	0.2206	0.2177	0.2148
−0.6	0.2743	0.2709	0.2676	0.2643	0.2611	0.2578	0.2546	0.2514	0.2483	0.2451
−0.5	0.3085	0.3050	0.3015	0.2981	0.2946	0.2912	0.2877	0.2843	0.2810	0.2776
−0.4	0.3446	0.3409	0.3372	0.3336	0.3300	0.3264	0.3228	0.3192	0.3156	0.3121
−0.3	0.3821	0.3783	0.3745	0.3707	0.3669	0.3632	0.3594	0.3557	0.3520	0.3483
−0.2	0.4207	0.4168	0.4129	0.4090	0.4052	0.4013	0.3974	0.3936	0.3897	0.3859
−0.1	0.4602	0.4562	0.4522	0.4483	0.4443	0.4404	0.4364	0.4325	0.4286	0.4247
−0.0	0.5000	0.4960	0.4920	0.4880	0.4840	0.4801	0.4761	0.4721	0.4681	0.4641

Standard Normal (Z) Table
Area between 0 and z



	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
0.0	0.0000	0.0040	0.0080	0.0120	0.0160	0.0199	0.0239	0.0279	0.0319	0.0359
0.1	0.0398	0.0438	0.0478	0.0517	0.0557	0.0596	0.0636	0.0675	0.0714	0.0753
0.2	0.0793	0.0832	0.0871	0.0910	0.0948	0.0987	0.1026	0.1064	0.1103	0.1141
0.3	0.1179	0.1217	0.1255	0.1293	0.1331	0.1368	0.1406	0.1443	0.1480	0.1517
0.4	0.1554	0.1591	0.1628	0.1664	0.1700	0.1736	0.1772	0.1808	0.1844	0.1879
0.5	0.1915	0.1950	0.1985	0.2019	0.2054	0.2088	0.2123	0.2157	0.2190	0.2224
0.6	0.2257	0.2291	0.2324	0.2357	0.2389	0.2422	0.2454	0.2486	0.2517	0.2549
0.7	0.2580	0.2611	0.2642	0.2673	0.2704	0.2734	0.2764	0.2794	0.2823	0.2852
0.8	0.2881	0.2910	0.2939	0.2967	0.2995	0.3023	0.3051	0.3078	0.3106	0.3133
0.9	0.3159	0.3186	0.3212	0.3238	0.3264	0.3289	0.3315	0.3340	0.3365	0.3389
1.0	0.3413	0.3438	0.3461	0.3485	0.3508	0.3531	0.3554	0.3577	0.3599	0.3621
1.1	0.3643	0.3665	0.3686	0.3708	0.3729	0.3749	0.3770	0.3790	0.3810	0.3830
1.2	0.3849	0.3869	0.3888	0.3907	0.3925	0.3944	0.3962	0.3980	0.3997	0.4015
1.3	0.4032	0.4049	0.4066	0.4082	0.4099	0.4115	0.4131	0.4147	0.4162	0.4177
1.4	0.4192	0.4207	0.4222	0.4236	0.4251	0.4265	0.4279	0.4292	0.4306	0.4319
1.5	0.4332	0.4345	0.4357	0.4370	0.4382	0.4394	0.4406	0.4418	0.4429	0.4441
1.6	0.4452	0.4463	0.4474	0.4484	0.4495	0.4505	0.4515	0.4525	0.4535	0.4545
1.7	0.4554	0.4564	0.4573	0.4582	0.4591	0.4599	0.4608	0.4616	0.4625	0.4633
1.8	0.4641	0.4649	0.4656	0.4664	0.4671	0.4678	0.4686	0.4693	0.4699	0.4706
1.9	0.4713	0.4719	0.4726	0.4732	0.4738	0.4744	0.4750	0.4756	0.4761	0.4767
2.0	0.4772	0.4778	0.4783	0.4788	0.4793	0.4798	0.4803	0.4808	0.4812	0.4817
2.1	0.4821	0.4826	0.4830	0.4834	0.4838	0.4842	0.4846	0.4850	0.4854	0.4857
2.2	0.4861	0.4864	0.4868	0.4871	0.4875	0.4878	0.4881	0.4884	0.4887	0.4890
2.3	0.4893	0.4896	0.4898	0.4901	0.4904	0.4906	0.4909	0.4911	0.4913	0.4916
2.4	0.4918	0.4920	0.4922	0.4925	0.4927	0.4929	0.4931	0.4932	0.4934	0.4936
2.5	0.4938	0.4940	0.4941	0.4943	0.4945	0.4946	0.4948	0.4949	0.4951	0.4952
2.6	0.4953	0.4955	0.4956	0.4957	0.4959	0.4960	0.4961	0.4962	0.4963	0.4964
2.7	0.4965	0.4966	0.4967	0.4968	0.4969	0.4970	0.4971	0.4972	0.4973	0.4974
2.8	0.4974	0.4975	0.4976	0.4977	0.4977	0.4978	0.4979	0.4979	0.4980	0.4981
2.9	0.4981	0.4982	0.4982	0.4983	0.4984	0.4984	0.4985	0.4985	0.4986	0.4986
3.0	0.4987	0.4987	0.4987	0.4988	0.4988	0.4989	0.4989	0.4989	0.4990	0.4990